

# Ahmed Nesar Tahsin Choudhury

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[GitHub](#) | [Codeforces](#) | [Google Scholar](#)

## Summary

Final-year Computer Science and Engineering undergraduate at the University of Dhaka, passionate about deep learning research, with a recent publication at EMNLP 2025 System Demonstrations, strong algorithmic and problem-solving skills from competitive programming

## Education

**University of Dhaka**, BSc in Computer Science and Engineering 2022 – 2026

- **CGPA:** 3.96/4.00 (till 4th Year, 1st Semester)
- **Rank:** 1 out of 65
- **Admission Test:** 59th out of ~90,000 applicants (**99.9th** Percentile)
- **Relevant Coursework:** Machine Learning, Artificial Intelligence, Mathematical and Statistical Analysis for Engineers, Introduction to Probability and Statistics, Introduction to Numerical Methods, Design and Analysis of Algorithms, Data Structures and Algorithms, Introduction to Software Engineering

## Experience

**Research Assistant**, Cognitive Agents and Interaction Lab, University of Dhaka June 2024 – Present

- Developed a state-of-the-art Bengali Handwritten Text Recognition System
- Experimented with encoder-decoder and decoder-only architectures in OCR
- Experimented with grapheme-level tokenization as an alternative to byte-level tokenization for complex scripts
- Showed that integrating linguistic context into tokenizers improves text recognition accuracy

**Software Engineer Intern**, Mass Holdings Inc. – Tokyo, Japan (Remote) July 2024 – Dec 2025

- Built a sales forecasting model with XGBoost to predict demand trends and improve decision-making
- Developed a small-scale live streaming prototype utilizing Nginx as the media server to process video streams through RTMP, delivering HLS segments
- Engineered a ticketing system featuring client ID generation and QR code verification for enhanced event management
- Designed a real-time interactive 3D object visualizer using smartphone cameras and Three.js, enabling clients to preview product fit within intended environments before purchasing
- Built foundational features for an e-commerce platform's backend, including carts, promotions, and order management, leveraging Spring Boot and MySQL

## Publications

[GraDeT-HTR: A Bengali Handwritten Text Recognition System utilizing Grapheme-based Tokenizer and Decoder-only Transformer](#) Sept 2025

Md. Mahmudul Hasan\*, **Ahmed Nesar Tahsin Choudhury\***, Mahmudul Hasan, Md. Mosaddek Khan

**Published** at 2025 Conference on Empirical Methods in Natural Language Processing: System Demonstrations

**Deployment:** <https://cognistorm.ai/hcr>

\* indicates equal contribution

## Projects

**Identification of Fungal Pathogens in Teak** June 2025

- Applied deep learning by experimenting with both a custom CNN architecture and a pre-trained ResNet model to classify fungal pathogens from microscopic teak images

- Utilized Python with PyTorch for model development, training, and evaluation
- Performed image preprocessing and augmentation using OpenCV to improve generalization and dataset robustness

### Neural Network

January 2025

- Implemented a foundational neural network entirely from scratch using NumPy for vectorized operations and pandas for dataset preprocessing
- Built forward and backward propagation manually to deepen understanding of gradient-based optimization
- Evaluated model performance using custom accuracy and loss metrics implemented in Python

### Locato

Sept 2024 – Nov 2024

- Developed a cross-platform Android application using React Native for responsive UI, smooth navigation, and device compatibility
- Engineered backend services using Spring Boot, enabling secure authentication and real-time communication between users and providers
- Integrated large language models (LLMs) to parse natural-language issue descriptions and intelligently match users with relevant service providers
- Utilized REST APIs to streamline data exchange between React Native frontend and Spring Boot backend

### GadgetGrove

March 2024 – May 2024

- Created an e-commerce platform using Spring Boot for backend services, including product management, user authentication, and order processing
- Built the frontend using Next.js
- Designed a responsive UI using Tailwind CSS to ensure seamless experience across devices
- Integrated PostgreSQL for reliable data storage and retrieval

### CoRide

Sept 2023 – Nov 2023

- Built an Android application in Java that enables motorbike riders to reduce travel costs through shared-route passenger matching
- Implemented Firebase Realtime Database for storing user trips and managing ride requests

### Carrom

Sept 2022 – Nov 2022

- Developed a desktop Carrom game for Linux using C++ with the SDL framework for graphics rendering and event handling
- Implemented realistic physics calculations for striker and coin movement
- Designed modular game logic to support future feature additions such as multiplayer modes

## Awards and Achievements

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- **2024: Runner-up**, CODE Samurai (National Hackathon)
- **2023:**
  - **26th** out of 196 teams in **NCPC Onsite 2023**, National Collegiate Programming Contest, JU
  - **24th** out of 96 teams in **CUET IUPC 2023**, Inter University Programming Contest 2023, CUET
  - **Champion** among 100 participants, Battle of Brains, University of Dhaka
- **2022:**
  - **Champion** (among freshmen), Battle of Brains, University of Dhaka
  - **45th** out of **1700+** teams, ICPC Dhaka Regionals Preliminary Round
- **2021: 59th** out of **~90,000 (99.9th percentile)** applicants, Admission Test – A unit, University of Dhaka
- **2020: General Board Scholarship Recipient**, HSC
- **2020: 8th** out of 393 participants in NHSPC, **National Camper**, National High School Programming Contest
- **2019: 11th** out of 1000+ participants in BDPhO, **National Camper**, Bangladesh Physics Olympiad
- **2018: Talentpool Board Scholarship Recipient**, SSC

- **2015: Talentpool Board Scholarship Recipient, JSC**

## **Skills**

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**Competitive Programming:** Codeforces Max Rating: 1588 (Username: Philoomath)

**ML/DL:** NumPy, Pandas, PyTorch

**Coding:** Java, Python, C, C++ , JavaScript, SQL, ~~W~~TeX, ...

**Databases:** MySQL, PostgreSQL

**Web Development:** Spring Boot, FastAPI, HTML, CSS, React JS (basics), ...

**Miscellaneous:** Tutoring high-school and college students

**Languages:** Bangla, English

## **References**

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**Dr. Md. Mosaddek Khan**

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